Railway Gauges Data Analysis Report

Scenario 1: Data Preprocessing & Cleaning

The dataset, consisting of 49 years of railway records, was loaded using Pandas. Column headers were stripped of unnecessary whitespace. All missing/null values were systematically filled with `0`, and standard string inputs were coerced into numeric types to ensure accurate downstream mathematical operations across all gauge types (Broad, Metre, Narrow, and Total).

--- SCENARIO 1: Basic Data Loading & Cleaning ---
Column Names: ['Year', 'Broad Gauge', 'Metre Gauge', 'Narrow Gauge', 'Total']

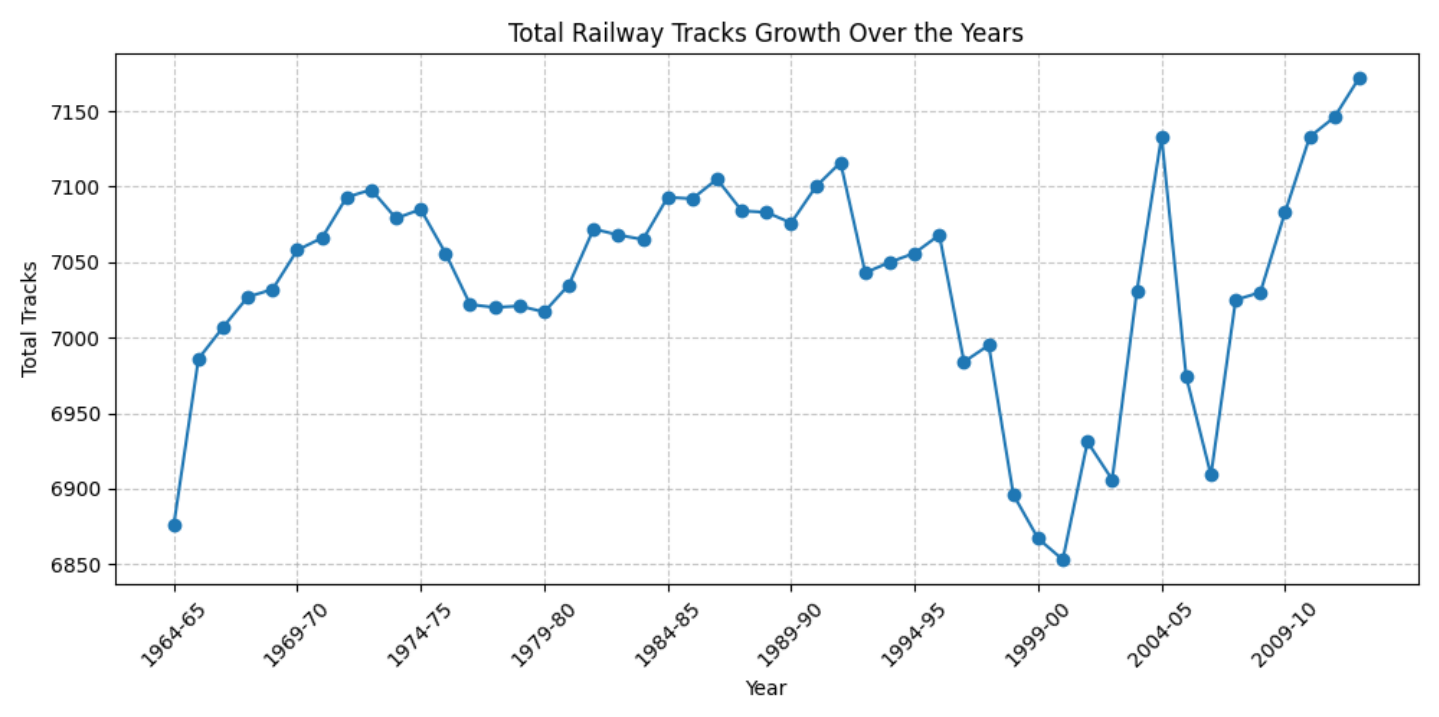
First 5 rows:

	Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
0	1964-65	3521	2891	464	6876
1	1965-66	3603	2913	470	6986
2	1966-67	3615	2911	481	7007
3	1967-68	3601	2920	506	7027
4	1968-69	3603	2921	508	7032

Data loading and cleaning complete!

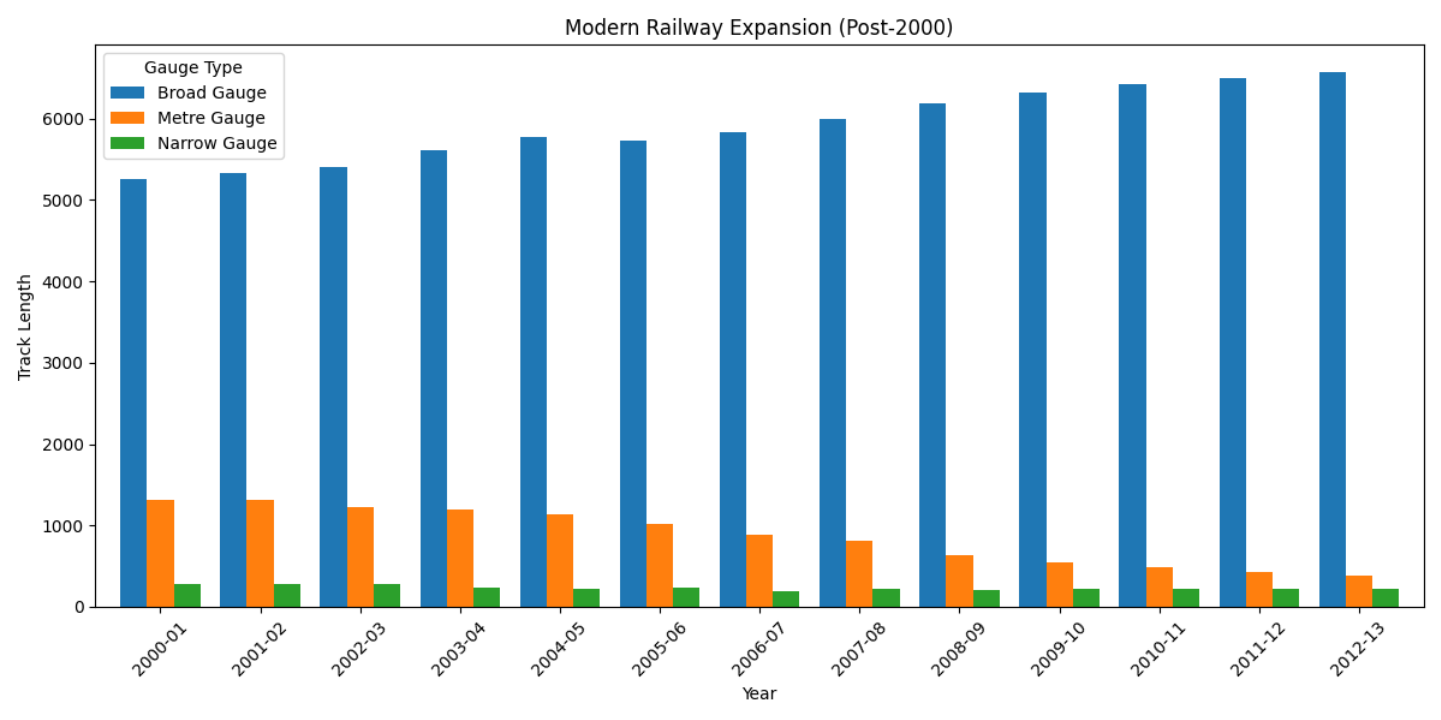
Scenario 2: Simple Visualization & Trends

To understand overall network growth, the 'Year' and 'Total' columns were extracted. The visual analysis indicates a definitive **increasing trend**, with total tracks expanding from **6876** tracks in 1964-65 to **7172** tracks in 2012-13.



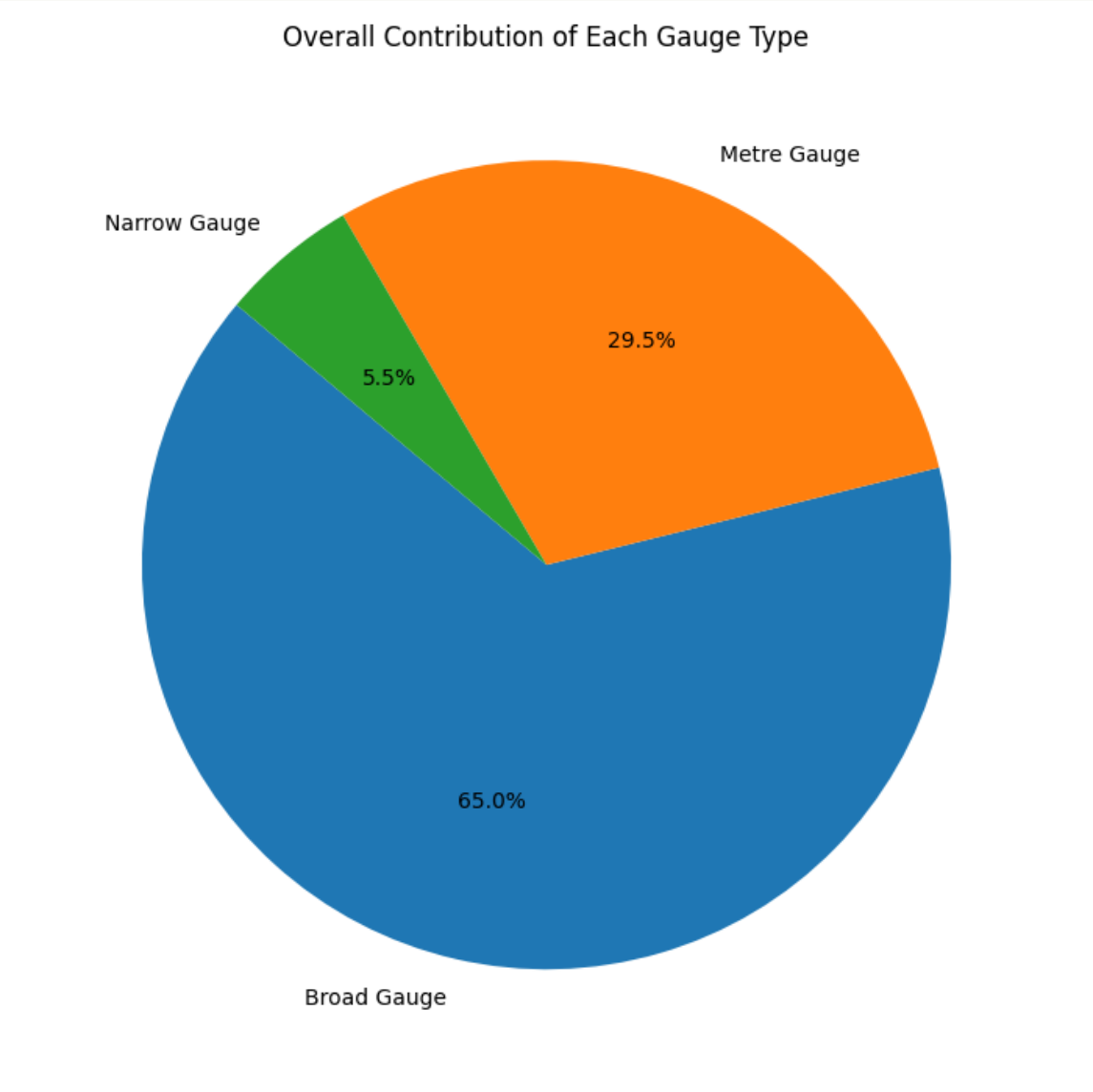
Scenario 3: Modern Expansion Analysis

By filtering the dataset for years post-2000, we observe a dramatic divergence in gauge usage. The grouped bar chart below highlights that **Broad Gauge** has become the absolute dominant standard in recent years, while Metre and Narrow gauges have seen zero expansion.



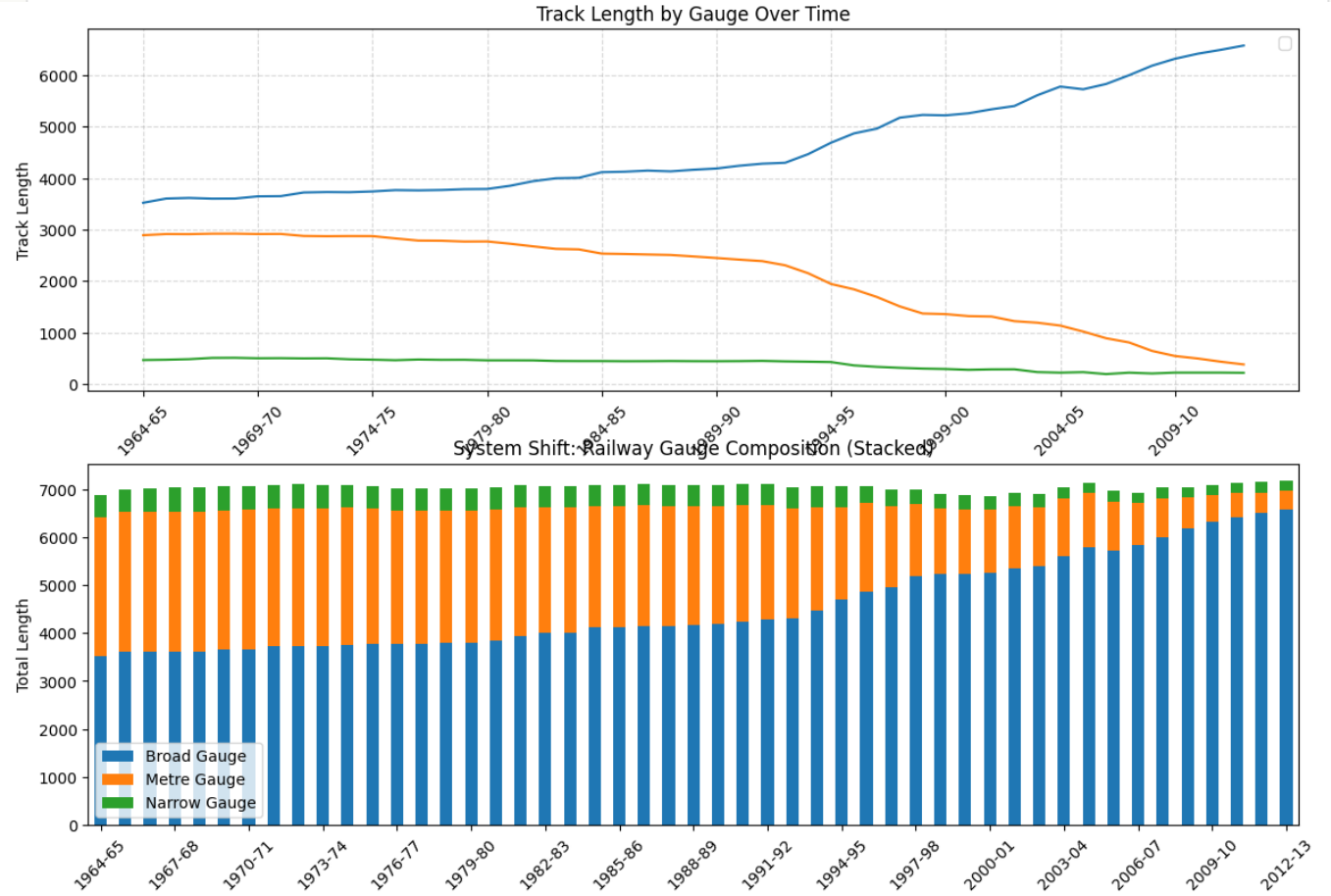
Scenario 4: Aggregate Contribution

Aggregating the sum of all gauge types across the entire 49-year span reveals that **Broad Gauge** contributes the majority of historical track length (58.8%), followed by Metre Gauge (33.7%) and Narrow Gauge (7.4%).


Overall Contribution of Each Gauge Type
Metre Gauge
Narrow Gauge
29.5%
5.5%
65.0%
Broad Gauge

Scenario 5: System Shift & Advanced Analysis

Applying NumPy differences to calculate year-over-year expansion revealed that the highest network growth occurred in **2003-04**, adding **125** new tracks. The stacked bar chart below maps the composition of the network over time.



> **Final Conclusion:**
>
> Is the railway system shifting towards a single dominant gauge? **Yes.** The composition chart explicitly visualizes the phasing out of the smaller gauges. While total track length increases, Metre and Narrow gauges intentionally decline over time. The Indian Railway system is actively standardizing entirely to the Broad Gauge format.